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By - SYNTAX ERROR

Step 4 : Binary Search

The PDF created by Abhishek Rathor (Instagram:SYNTAX ERROR)

PROBLEM: 1

Problem Statement:

Write a Java program to perform Binary Search to find x in a sorted array.

Java Code:

```
java
CopyEdit
import java.util.Scanner;

public class BinarySearch {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter size of array: ");
        int n = scanner.nextInt();
        int[] arr = new int[n];

        System.out.println("Enter sorted array elements: ");
        for (int i = 0; i < n; i++) {
            arr[i] = scanner.nextInt();
        }

        System.out.print("Enter the number to search: ");
        int X = scanner.nextInt();
        scanner.close();

        int result = search(arr, X);
        System.out.println("Index of " + X + " in array: " + result);
    }

    public static int search(int[] arr, int X) {
        int left = 0, right = arr.length - 1;

        while (left <= right) {
            int mid = left + (right - left) / 2;

            if (arr[mid] == X) return mid;
            else if (arr[mid] < X) left = mid + 1;
            else right = mid - 1;
        }

        return -1;
    }
}
```

Explanation:

1. The program takes the size of a sorted array and elements as input from the user.
2. The user enters the value x that needs to be searched.
3. The `search` method implements **Binary Search** to find x in the array.
4. If x is found, it returns its index; otherwise, it returns -1 .
5. The result is printed to the console.

PROBLEM: 2

Problem Statement:

Write a Java program to implement **Lower Bound** of x in a sorted array.

Java Code:

```
java
CopyEdit
import java.util.Scanner;

public class LowerBound {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter size of array: ");
        int n = scanner.nextInt();
        int[] arr = new int[n];

        System.out.println("Enter sorted array elements: ");
        for (int i = 0; i < n; i++) {
            arr[i] = scanner.nextInt();
        }

        System.out.print("Enter the number to find lower bound: ");
        int X = scanner.nextInt();
        scanner.close();

        int result = lowerBound(arr, X);
        System.out.println("Lower bound index of " + X + ": " + result);
    }

    public static int lowerBound(int[] arr, int X) {
        int left = 0, right = arr.length;

        while (left < right) {
            int mid = left + (right - left) / 2;

            if (arr[mid] >= X) right = mid;
            else left = mid + 1;
        }

        return left;
    }
}
```

Explanation:

1. The program takes a sorted array and a number x as input.
2. The `lowerBound` function finds the index of the first element **greater than or equal to** x .
3. If x exists in the array, it returns its first occurrence index.
4. If x is greater than all elements, it returns the array length.
5. The index is printed as output.

PROBLEM: 3

Problem Statement:

Write a Java program to implement **Upper Bound** of x in a sorted array.

Java Code:

```
java
CopyEdit
import java.util.Scanner;

public class UpperBound {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter size of array: ");
        int n = scanner.nextInt();
        int[] arr = new int[n];

        System.out.println("Enter sorted array elements: ");
        for (int i = 0; i < n; i++) {
            arr[i] = scanner.nextInt();
        }

        System.out.print("Enter the number to find upper bound: ");
        int X = scanner.nextInt();
        scanner.close();

        int result = upperBound(arr, X);
        System.out.println("Upper bound index of " + X + ": " + result);
    }

    public static int upperBound(int[] arr, int X) {
        int left = 0, right = arr.length;

        while (left < right) {
            int mid = left + (right - left) / 2;

            if (arr[mid] > X) right = mid;
            else left = mid + 1;
        }

        return left;
    }
}
```

Explanation:

1. The program takes a sorted array and a number x as input.
2. The `upperBound` function finds the index of the first element **strictly greater than** x .
3. If x exists multiple times, it returns the index just after the last occurrence.
4. The index is printed as output.

PROBLEM: 4

Problem Statement:

Write a Java program to find the **Search Insert Position** of x in a sorted array.

Java Code:

```
java
CopyEdit
import java.util.Scanner;

public class SearchInsertPosition {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter size of array: ");
        int n = scanner.nextInt();
        int[] arr = new int[n];

        System.out.println("Enter sorted array elements: ");
        for (int i = 0; i < n; i++) {
            arr[i] = scanner.nextInt();
        }

        System.out.print("Enter the number to find insert position: ");
        int X = scanner.nextInt();
        scanner.close();

        int result = searchInsert(arr, X);
        System.out.println("Insert position of " + X + ": " + result);
    }

    public static int searchInsert(int[] arr, int X) {
        int left = 0, right = arr.length - 1;

        while (left <= right) {
            int mid = left + (right - left) / 2;

            if (arr[mid] == X) return mid;
            else if (arr[mid] < X) left = mid + 1;
            else right = mid - 1;
        }

        return left;
    }
}
```

Explanation:

1. The program takes a sorted array and an integer x as input.
2. It uses Binary Search to determine the **exact position** where x should be inserted to maintain sorted order.
3. If x is found, it returns its index.
4. Otherwise, it returns the position where x should be inserted.

PROBLEM: 5

Problem Statement:

Write a Java program to find **Floor and Ceil** of x in a sorted array.

Java Code:

```
java
CopyEdit
import java.util.Scanner;

public class FloorCeil {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter size of array: ");
        int n = scanner.nextInt();
        int[] arr = new int[n];

        System.out.println("Enter sorted array elements: ");
        for (int i = 0; i < n; i++) {
            arr[i] = scanner.nextInt();
        }

        System.out.print("Enter the number to find Floor and Ceil: ");
        int X = scanner.nextInt();
        scanner.close();

        int floor = findFloor(arr, X);
        int ceil = findCeil(arr, X);

        System.out.println("Floor of " + X + ": " + (floor == -1 ? "None" :
floor));
        System.out.println("Ceil of " + X + ": " + (ceil == -1 ? "None" :
ceil));
    }

    public static int findFloor(int[] arr, int X) {
        int left = 0, right = arr.length - 1, floor = -1;

        while (left <= right) {
            int mid = left + (right - left) / 2;

            if (arr[mid] <= X) {
                floor = arr[mid];
                left = mid + 1;
            } else {
                right = mid - 1;
            }
        }
    }
}
```

```

        }
    }

    return floor;
}

public static int findCeil(int[] arr, int X) {
    int left = 0, right = arr.length - 1, ceil = -1;

    while (left <= right) {
        int mid = left + (right - left) / 2;

        if (arr[mid] >= X) {
            ceil = arr[mid];
            right = mid - 1;
        } else {
            left = mid + 1;
        }
    }

    return ceil;
}
}

```

Explanation:

1. The program takes a sorted array and a number x.
 2. findFloor function finds the largest number $\leq x$ (Floor).
 3. findCeil function finds the smallest number $\geq x$ (Ceil).
 4. If no floor or ceil exists, it returns -1.
-

PROBLEM: 6

Problem Statement:

Write a Java program to find the **First and Last occurrence** of x in a sorted array.

Java Code:

```

java
CopyEdit
import java.util.Scanner;

public class FirstLastOccurrence {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter size of array: ");
        int n = scanner.nextInt();
        int[] arr = new int[n];

        System.out.println("Enter sorted array elements: ");
        for (int i = 0; i < n; i++) {

```

```

        arr[i] = scanner.nextInt();
    }

    System.out.print("Enter the number to find occurrences: ");
    int X = scanner.nextInt();
    scanner.close();

    int first = findFirst(arr, X);
    int last = findLast(arr, X);

    System.out.println("First occurrence of " + X + ": " + first);
    System.out.println("Last occurrence of " + X + ": " + last);
}

public static int findFirst(int[] arr, int X) {
    int left = 0, right = arr.length - 1, first = -1;

    while (left <= right) {
        int mid = left + (right - left) / 2;

        if (arr[mid] == X) {
            first = mid;
            right = mid - 1;
        } else if (arr[mid] < X) {
            left = mid + 1;
        } else {
            right = mid - 1;
        }
    }

    return first;
}

public static int findLast(int[] arr, int X) {
    int left = 0, right = arr.length - 1, last = -1;

    while (left <= right) {
        int mid = left + (right - left) / 2;

        if (arr[mid] == X) {
            last = mid;
            left = mid + 1;
        } else if (arr[mid] < X) {
            left = mid + 1;
        } else {
            right = mid - 1;
        }
    }

    return last;
}
}

```

Explanation:

1. The program takes a sorted array and a number x.
2. `findFirst` function finds the first occurrence of x in the array.
3. `findLast` function finds the last occurrence of x in the array.

4. If x is not found, it returns -1.

PROBLEM: 7

Problem Statement:

Write a Java program to count occurrences of a given number in a sorted array efficiently.

Java Code:

```
java
CopyEdit
import java.util.Scanner;

public class CountOccurrences {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter size of array: ");
        int n = scanner.nextInt();
        int[] arr = new int[n];

        System.out.println("Enter sorted array elements: ");
        for (int i = 0; i < n; i++) {
            arr[i] = scanner.nextInt();
        }

        System.out.print("Enter the number to count occurrences: ");
        int X = scanner.nextInt();
        scanner.close();

        int result = countOccurrences(arr, X);
        System.out.println("Occurrences of " + X + ": " + result);
    }

    public static int countOccurrences(int[] arr, int X) {
        int first = firstOccurrence(arr, X);
        if (first == -1) return 0;
        int last = lastOccurrence(arr, X);
        return last - first + 1;
    }

    public static int firstOccurrence(int[] arr, int X) {
        int left = 0, right = arr.length - 1, ans = -1;
        while (left <= right) {
            int mid = left + (right - left) / 2;
            if (arr[mid] == X) {
                ans = mid;
                right = mid - 1;
            } else if (arr[mid] < X) left = mid + 1;
            else right = mid - 1;
        }
        return ans;
    }

    public static int lastOccurrence(int[] arr, int X) {
        int left = 0, right = arr.length - 1, ans = -1;
        while (left <= right) {
            int mid = left + (right - left) / 2;
            if (arr[mid] == X) {
                ans = mid;
```

```

        left = mid + 1;
    } else if (arr[mid] < X) left = mid + 1;
    else right = mid - 1;
}
return ans;
}
}

```

Explanation:

1. Uses binary search to find the **first** and **last occurrence** of x in the sorted array.
 2. If x is not found, return 0.
 3. Else, count occurrences as `last - first + 1`.
-

PROBLEM: 8

Problem Statement:

Write a Java program to search for a target in a rotated sorted array.

Java Code:

```

java
CopyEdit
import java.util.Scanner;

public class RotatedArraySearch {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter size of array: ");
        int n = scanner.nextInt();
        int[] arr = new int[n];

        System.out.println("Enter rotated sorted array elements: ");
        for (int i = 0; i < n; i++) {
            arr[i] = scanner.nextInt();
        }

        System.out.print("Enter the number to search: ");
        int target = scanner.nextInt();
        scanner.close();

        int result = search(arr, target);
        System.out.println("Index of " + target + ": " + result);
    }

    public static int search(int[] arr, int target) {
        int left = 0, right = arr.length - 1;

        while (left <= right) {
            int mid = left + (right - left) / 2;

            if (arr[mid] == target) return mid;

            if (arr[left] <= arr[mid]) {

```

```

        if (arr[left] <= target && target < arr[mid]) right = mid -
1;
        else left = mid + 1;
    } else {
        if (arr[mid] < target && target <= arr[right]) left = mid +
1;
        else right = mid - 1;
    }
}

return -1;
}
}

```

Explanation:

1. Uses binary search but considers the rotated property.
 2. Determines which half is sorted and decides where to search next.
 3. Time Complexity: **$O(\log N)$** .
-

PROBLEM: 9

Problem Statement:

Write a Java program to find a peak element in an array, where an element is a peak if it is greater than its neighbors.

Java Code:

```

java
CopyEdit
import java.util.Scanner;

public class PeakElement {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter size of array: ");
        int n = scanner.nextInt();
        int[] arr = new int[n];

        System.out.println("Enter array elements: ");
        for (int i = 0; i < n; i++) {
            arr[i] = scanner.nextInt();
        }
        scanner.close();

        int result = findPeak(arr);
        System.out.println("Peak element is at index: " + result);
    }

    public static int findPeak(int[] arr) {
        int left = 0, right = arr.length - 1;

        while (left < right) {
            int mid = left + (right - left) / 2;

```

```

        if (arr[mid] > arr[mid + 1]) {
            right = mid;
        } else {
            left = mid + 1;
        }
    }

    return left;
}
}

```

PROBLEM: 10

Explanation:

1. Uses **binary search** to locate a peak element efficiently.
2. If $\text{arr}[\text{mid}] > \text{arr}[\text{mid}+1]$, move left. Otherwise, move right.
3. Time Complexity: **$O(\log N)$** .

Problem Statement:

Write a Java program to search for a target in a rotated sorted array that may contain duplicate values.

Java Code:

```

java
CopyEdit
import java.util.Scanner;

public class RotatedArraySearchII {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter size of array: ");
        int n = scanner.nextInt();
        int[] arr = new int[n];

        System.out.println("Enter rotated sorted array elements (may
contain duplicates): ");
        for (int i = 0; i < n; i++) {
            arr[i] = scanner.nextInt();
        }

        System.out.print("Enter the number to search: ");
        int target = scanner.nextInt();
        scanner.close();

        boolean result = search(arr, target);
        System.out.println("Is " + target + " present? " + result);
    }

    public static boolean search(int[] arr, int target) {
        int left = 0, right = arr.length - 1;
    }
}

```

```

while (left <= right) {
    int mid = left + (right - left) / 2;

    if (arr[mid] == target) return true;

    if (arr[left] == arr[mid] && arr[mid] == arr[right]) {
        left++;
        right--;
    } else if (arr[left] <= arr[mid]) {
        if (arr[left] <= target && target < arr[mid]) right = mid -
1;
        else left = mid + 1;
    } else {
        if (arr[mid] < target && target <= arr[right]) left = mid +
1;
        else right = mid - 1;
    }
}

return false;
}
}

```

Explanation:

1. Similar to **Search in Rotated Sorted Array I**, but handles **duplicates**.
 2. If `arr[left] == arr[mid] == arr[right]`, increment `left` and decrement `right` to skip duplicates.
 3. Uses **binary search** for **$O(\log N)$ worst-case $O(N)$ complexity**.
-

PROBLEM: 11

Problem Statement:

Write a Java program to find the minimum element in a rotated sorted array.

Java Code:

```

java
CopyEdit
import java.util.Scanner;

public class MinInRotatedArray {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter size of array: ");
        int n = scanner.nextInt();
        int[] arr = new int[n];

        System.out.println("Enter rotated sorted array elements: ");
        for (int i = 0; i < n; i++) {
            arr[i] = scanner.nextInt();
        }
        scanner.close();
    }
}

```

```

        int result = findMin(arr);
        System.out.println("Minimum element in rotated array: " + result);
    }

    public static int findMin(int[] arr) {
        int left = 0, right = arr.length - 1;

        while (left < right) {
            int mid = left + (right - left) / 2;

            if (arr[mid] > arr[right]) left = mid + 1;
            else right = mid;
        }

        return arr[left];
    }
}

```

Explanation:

1. Uses **binary search** to find the **pivot** point.
 2. If `arr[mid] > arr[right]`, then **min is on the right half**.
 3. Else, **min is on the left half**.
 4. Runs in **$O(\log N)$ time**.
-

PROBLEM: 12

Problem Statement:

Write a Java program to determine how many times a sorted array has been rotated.

Java Code:

```

java
CopyEdit
import java.util.Scanner;

public class RotationCount {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter size of array: ");
        int n = scanner.nextInt();
        int[] arr = new int[n];

        System.out.println("Enter rotated sorted array elements: ");
        for (int i = 0; i < n; i++) {
            arr[i] = scanner.nextInt();
        }
        scanner.close();

        int result = rotationCount(arr);
        System.out.println("Array is rotated " + result + " times.");
    }
}

```

```

public static int rotationCount(int[] arr) {
    int left = 0, right = arr.length - 1;

    while (left < right) {
        int mid = left + (right - left) / 2;

        if (arr[mid] > arr[right]) left = mid + 1;
        else right = mid;
    }

    return left;
}
}

```

Explanation:

1. Finds **minimum element's index**, which is the **number of rotations**.
 2. Works the same way as **Find Minimum in Rotated Array**.
 3. Time Complexity: **$O(\log N)$** .
-

PROBLEM: 13

Problem Statement:

Write a Java program to find the **single non-repeating** element in a sorted array where every other element appears twice.

Java Code:

```

java
CopyEdit
import java.util.Scanner;

public class SingleElement {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter size of array: ");
        int n = scanner.nextInt();
        int[] arr = new int[n];

        System.out.println("Enter sorted array elements (every number
appears twice except one): ");
        for (int i = 0; i < n; i++) {
            arr[i] = scanner.nextInt();
        }
        scanner.close();

        int result = findSingleElement(arr);
        System.out.println("Single non-repeating element: " + result);
    }

    public static int findSingleElement(int[] arr) {
        int left = 0, right = arr.length - 1;

        while (left < right) {

```

```

        int mid = left + (right - left) / 2;
        if (mid % 2 == 1) mid--;
        if (arr[mid] == arr[mid + 1]) left = mid + 2;
        else right = mid;
    }

    return arr[left];
}
}

```

Explanation:

1. Uses **binary search** with **index-based checking**.
2. If `arr[mid] == arr[mid + 1]`, move right. Otherwise, move left.
3. Time Complexity: **O(log N)**.

PROBLEM: 14

Problem Statement:

Write a Java program to find a **peak element** in an array. A peak element is an element that is strictly greater than its neighbors.

Java Code:

```

java
CopyEdit
import java.util.Scanner;

public class PeakElement {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter size of array: ");
        int n = scanner.nextInt();
        int[] arr = new int[n];

        System.out.println("Enter array elements: ");
        for (int i = 0; i < n; i++) {
            arr[i] = scanner.nextInt();
        }
        scanner.close();

        int peakIndex = findPeakElement(arr);
        System.out.println("Peak element is at index: " + peakIndex + "
(Value: " + arr[peakIndex] + ")");
    }

    public static int findPeakElement(int[] arr) {
        int left = 0, right = arr.length - 1;

        while (left < right) {
            int mid = left + (right - left) / 2;

            if (arr[mid] > arr[mid + 1]) right = mid;
            else left = mid + 1;
        }
    }
}

```



```
    }  
    return left;  
}  
}
```

Explanation:

1. Uses **binary search** to find a peak element.
2. If `arr[mid] > arr[mid + 1]`, the **peak lies in the left half**.
3. Otherwise, the **peak lies in the right half**.
4. Time Complexity: **$O(\log N)$** .

PROBLEM: 15

Problem Statement:

Write a Java program to find the **integer square root** of a given number using **binary search** in **$O(\log N)$** time complexity.

Java Code:

```
java  
CopyEdit  
import java.util.Scanner;  
  
public class SquareRootBinarySearch {  
    public static void main(String[] args) {  
        Scanner scanner = new Scanner(System.in);  
        System.out.print("Enter a number: ");  
        int num = scanner.nextInt();  
        scanner.close();  
  
        int sqrt = findSquareRoot(num);  
        System.out.println("Square root of " + num + " is: " + sqrt);  
    }  
  
    public static int findSquareRoot(int num) {  
        if (num == 0 || num == 1) return num;  
  
        int left = 1, right = num, ans = 0;  
        while (left <= right) {  
            int mid = left + (right - left) / 2;  
            if (mid * mid == num) return mid;  
            if (mid * mid < num) {  
                ans = mid;  
                left = mid + 1;  
            } else {  
                right = mid - 1;  
            }  
        }  
        return ans;  
    }  
}
```

Explanation:

1. We use **binary search** between 1 and num.
2. If $mid * mid == num$, return mid (**perfect square**).
3. If $mid * mid < num$, move **right** ($left = mid + 1$), store ans.
4. If $mid * mid > num$, move **left** ($right = mid - 1$).
5. Returns **integer square root** (floor value).

Example:

```
vbnet
CopyEdit
Input:  num = 17
Output: Square root of 17 is: 4
```

 **Time Complexity: $O(\log N)$.** Efficient for large numbers!

PROBLEM: 16

Problem Statement:

Write a Java program to find the **N-th root** of a number using **binary search**.

Java Code:

```
java
CopyEdit
import java.util.Scanner;

public class NthRootBinarySearch {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter the number: ");
        int num = scanner.nextInt();
        System.out.print("Enter the value of N: ");
        int n = scanner.nextInt();
        scanner.close();

        double nthRoot = findNthRoot(num, n);
        System.out.println("The " + n + "th root of " + num + " is: " +
            nthRoot);
    }

    public static double findNthRoot(int num, int n) {
        double low = 1, high = num, epsilon = 1e-6;
        while ((high - low) > epsilon) {
            double mid = (low + high) / 2.0;
            if (Math.pow(mid, n) < num) low = mid;
            else high = mid;
        }
        return high;
    }
}
```

Explanation:

1. Uses **binary search** in **floating point precision**.
2. If $\text{mid}^n < \text{num}$, move $\text{low} = \text{mid}$.
3. If $\text{mid}^n > \text{num}$, move $\text{high} = \text{mid}$.
4. Returns **nth root with precision** ($\text{epsilon} = 10^{-6}$).

Example:

```
vbnet
CopyEdit
Input:  num = 16, n = 4
Output: The 4th root of 16 is: 2.000000
```

 **Time Complexity: $O(\log N)$ using floating point binary search.**

PROBLEM: 17

Problem Statement:

Given `piles` of bananas and `h` hours, find the **minimum eating speed** (`k`) such that Koko eats all bananas within `h` hours.

Java Code:

```
java
CopyEdit
import java.util.Scanner;

public class KokoEatingBananas {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter number of piles: ");
        int n = scanner.nextInt();
        int[] piles = new int[n];

        System.out.println("Enter banana piles: ");
        for (int i = 0; i < n; i++) {
            piles[i] = scanner.nextInt();
        }

        System.out.print("Enter total hours (h): ");
        int h = scanner.nextInt();
        scanner.close();

        int minSpeed = minEatingSpeed(piles, h);
        System.out.println("Minimum eating speed: " + minSpeed);
    }

    public static int minEatingSpeed(int[] piles, int h) {
        int left = 1, right = Integer.MAX_VALUE;
```

```

        while (left < right) {
            int mid = left + (right - left) / 2;
            if (canEatAll(piles, mid, h)) right = mid;
            else left = mid + 1;
        }
        return left;
    }

    private static boolean canEatAll(int[] piles, int k, int h) {
        int hours = 0;
        for (int pile : piles) {
            hours += (pile + k - 1) / k;
        }
        return hours <= h;
    }
}

```

Explanation:

1. Uses **binary search** for k (eating speed).
 2. Check if Koko can **finish in h hours**.
 3. If `canEatAll(piles, k, h)`, decrease k .
 4. If not, increase k .
-

Example:

```

makefile
CopyEdit
Input:  piles = [3,6,7,11], h = 8
Output: Minimum eating speed: 4

```

 **Time Complexity:** $O(N \log M)$ (M is max pile size).

PROBLEM: 18

Problem Statement:

Find the **minimum number of days** to make m bouquets, each requiring k adjacent flowers to bloom.

Java Code:

```

java
CopyEdit
import java.util.Scanner;
import java.util.Arrays;

public class MinDaysBouquets {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter number of flowers: ");
    }
}

```

```

int n = scanner.nextInt();
int[] bloomDay = new int[n];

System.out.println("Enter bloom days: ");
for (int i = 0; i < n; i++) {
    bloomDay[i] = scanner.nextInt();
}

System.out.print("Enter bouquets needed: ");
int m = scanner.nextInt();
System.out.print("Enter flowers per bouquet: ");
int k = scanner.nextInt();
scanner.close();

int minDays = minDaysToMakeBouquets(bloomDay, m, k);
System.out.println("Minimum days to make " + m + " bouquets: " +
minDays);
}

public static int minDaysToMakeBouquets(int[] bloomDay, int m, int k) {
    if (bloomDay.length < m * k) return -1;

    int left = Arrays.stream(bloomDay).min().getAsInt();
    int right = Arrays.stream(bloomDay).max().getAsInt();

    while (left < right) {
        int mid = left + (right - left) / 2;
        if (canMakeBouquets(bloomDay, mid, m, k)) right = mid;
        else left = mid + 1;
    }
    return left;
}

private static boolean canMakeBouquets(int[] bloomDay, int days, int m,
int k) {
    int flowers = 0, bouquets = 0;
    for (int bloom : bloomDay) {
        if (bloom <= days) flowers++;
        else flowers = 0;

        if (flowers == k) {
            bouquets++;
            flowers = 0;
        }
    }
    return bouquets >= m;
}
}

```

Explanation:

1. Binary search for days.
2. Check if days can make m bouquets.
3. Adjust search range accordingly.

PROBLEM: 19

Problem Statement:

Given an array `nums` and a threshold `t`, find the **smallest divisor** such that the sum of all elements divided by it does not exceed `t`.

Java Code:

```
java
CopyEdit
import java.util.Scanner;

public class SmallestDivisor {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter array size: ");
        int n = scanner.nextInt();
        int[] nums = new int[n];

        System.out.println("Enter array elements: ");
        for (int i = 0; i < n; i++) {
            nums[i] = scanner.nextInt();
        }

        System.out.print("Enter threshold: ");
        int threshold = scanner.nextInt();
        scanner.close();

        int result = findSmallestDivisor(nums, threshold);
        System.out.println("Smallest Divisor: " + result);
    }

    public static int findSmallestDivisor(int[] nums, int threshold) {
        int left = 1, right = Integer.MAX_VALUE;
        while (left < right) {
            int mid = left + (right - left) / 2;
            if (sumDividedBy(nums, mid) <= threshold) right = mid;
            else left = mid + 1;
        }
        return left;
    }

    private static int sumDividedBy(int[] nums, int divisor) {
        int sum = 0;
        for (int num : nums) {
            sum += (num + divisor - 1) / divisor;
        }
        return sum;
    }
}
```

Explanation:

1. **Binary search** on possible divisors (1 to max element).
 2. If $\text{sum} \leq \text{threshold}$, decrease divisor ($\text{right} = \text{mid}$).
 3. Otherwise, increase divisor ($\text{left} = \text{mid} + 1$).
-

Example:

```
makefile
CopyEdit
Input: nums = [1,2,5,9], threshold = 6
Output: Smallest Divisor: 5
```

 **Time Complexity: $O(N \log M)$** (M is max element).

PROBLEM: 20

Problem Statement:

Given an array representing **weights of packages**, find the **minimum ship capacity** to ship them within D days.

Java Code:

```
java
CopyEdit
import java.util.Scanner;

public class ShipCapacity {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter number of packages: ");
        int n = scanner.nextInt();
        int[] weights = new int[n];

        System.out.println("Enter package weights: ");
        for (int i = 0; i < n; i++) {
            weights[i] = scanner.nextInt();
        }

        System.out.print("Enter number of days: ");
        int D = scanner.nextInt();
        scanner.close();

        int result = minShipCapacity(weights, D);
        System.out.println("Minimum Ship Capacity: " + result);
    }

    public static int minShipCapacity(int[] weights, int D) {
        int left = 0, right = 0;
        for (int weight : weights) {
            left = Math.max(left, weight);
            right += weight;
        }

        while (left < right) {
            int mid = left + (right - left) / 2;
            if (canShip(weights, D, mid)) right = mid;
            else left = mid + 1;
        }
        return left;
    }
}
```

```

private static boolean canShip(int[] weights, int D, int capacity) {
    int days = 1, load = 0;
    for (int weight : weights) {
        if (load + weight > capacity) {
            days++;
            load = 0;
        }
        load += weight;
    }
    return days <= D;
}
}

```

Explanation:

1. **Binary search** between $\max(\text{weight})$ (min capacity) and $\text{sum}(\text{weights})$ (max capacity).
2. Check if a **capacity** can ship in D days.
3. If `canShip()`, decrease capacity (`right = mid`).
4. Else, increase capacity (`left = mid + 1`).

Example:

```

makefile
CopyEdit
Input: weights = [1,2,3,4,5,6,7,8,9,10], D = 5
Output: Minimum Ship Capacity: 15

```

 **Time Complexity:** $O(N \log M)$ (M is sum of weights).

PROBLEM: 21

Problem Statement:

Find the **K-th missing positive number** from a sorted array of unique positive integers.

Java Code:

```

java
CopyEdit
import java.util.Scanner;

public class KthMissingPositive {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter array size: ");
        int n = scanner.nextInt();
        int[] arr = new int[n];

        System.out.println("Enter sorted array elements: ");
        for (int i = 0; i < n; i++) {
            arr[i] = scanner.nextInt();
        }
    }
}

```



```

    }

    System.out.print("Enter K: ");
    int k = scanner.nextInt();
    scanner.close();

    int result = findKthMissing(arr, k);
    System.out.println("Kth Missing Positive Number: " + result);
}

public static int findKthMissing(int[] arr, int k) {
    int left = 0, right = arr.length - 1;
    while (left <= right) {
        int mid = left + (right - left) / 2;
        int missing = arr[mid] - (mid + 1);
        if (missing < k) left = mid + 1;
        else right = mid - 1;
    }
    return left + k;
}
}

```

Explanation:

1. **Binary search** on $\text{arr}[i] - (i + 1)$.
 2. If missing numbers so far $< k$, increase left.
 3. Otherwise, decrease right.
 4. Final result is $\text{left} + k$.
-

Example:

```

mathematica
CopyEdit
Input: arr = [2,3,4,7,11], k = 5
Output: Kth Missing Positive Number: 9

```

 **Time Complexity: $O(\log N)$.**

PROBLEM: 22

Problem Statement:

Given n stalls and c cows, place the cows in the stalls so that the **minimum distance** between any two cows is **maximized**.

Java Code:

```

java
CopyEdit
import java.util.Arrays;
import java.util.Scanner;

public class AggressiveCows {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
    }
}

```

```

        System.out.print("Enter number of stalls: ");
        int n = scanner.nextInt();
        int[] stalls = new int[n];

        System.out.println("Enter stall positions: ");
        for (int i = 0; i < n; i++) {
            stalls[i] = scanner.nextInt();
        }

        System.out.print("Enter number of cows: ");
        int cows = scanner.nextInt();
        scanner.close();

        int result = findMaxMinDistance(stalls, cows);
        System.out.println("Largest minimum distance: " + result);
    }

    public static int findMaxMinDistance(int[] stalls, int cows) {
        Arrays.sort(stalls);
        int left = 1, right = stalls[n - 1] - stalls[0];
        int result = 0;

        while (left <= right) {
            int mid = left + (right - left) / 2;
            if (canPlaceCows(stalls, cows, mid)) {
                result = mid;
                left = mid + 1;
            } else {
                right = mid - 1;
            }
        }
        return result;
    }

    private static boolean canPlaceCows(int[] stalls, int cows, int minDist)
    {
        int count = 1, lastPlaced = stalls[0];
        for (int i = 1; i < stalls.length; i++) {
            if (stalls[i] - lastPlaced >= minDist) {
                count++;
                lastPlaced = stalls[i];
            }
            if (count == cows) return true;
        }
        return false;
    }
}

```

Explanation:

1. **Sort the stalls** in ascending order.
2. **Binary search** on possible distances.
3. Use `canPlaceCows()` to check if cows can be placed with `minDist`.
4. If yes, try a larger distance (`left = mid + 1`); otherwise, reduce (`right = mid - 1`).

Example:

makefile

CopyEdit
Input: stalls = [1, 2, 8, 4, 9], cows = 3
Output: Largest minimum distance: 3

 **Time Complexity: $O(N \log M)$** (M is max stall position).

PROBLEM: 23

Problem Statement:

Allocate n books to m students such that the **maximum pages assigned to a student** is minimized.

Java Code:

```
java
CopyEdit
import java.util.Scanner;

public class BookAllocation {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter number of books: ");
        int n = scanner.nextInt();
        int[] books = new int[n];

        System.out.println("Enter pages in books: ");
        for (int i = 0; i < n; i++) {
            books[i] = scanner.nextInt();
        }

        System.out.print("Enter number of students: ");
        int students = scanner.nextInt();
        scanner.close();

        int result = findMinimumPages(books, students);
        System.out.println("Minimum max pages allocated: " + result);
    }

    public static int findMinimumPages(int[] books, int students) {
        int left = 0, right = 0;
        for (int pages : books) {
            left = Math.max(left, pages);
            right += pages;
        }

        while (left < right) {
            int mid = left + (right - left) / 2;
            if (canAllocate(books, students, mid)) right = mid;
            else left = mid + 1;
        }
        return left;
    }

    private static boolean canAllocate(int[] books, int students, int
maxPages) {
        int studentCount = 1, pagesSum = 0;
```

```

        for (int pages : books) {
            if (pagesSum + pages > maxPages) {
                studentCount++;
                pagesSum = 0;
            }
            pagesSum += pages;
        }
        return studentCount <= students;
    }
}

```

Explanation:

1. **Binary search** between `max(books)` and `sum(books)`.
 2. Check if `mid` pages can be allocated using `canAllocate()`.
 3. If `canAllocate()`, try a smaller value (`right = mid`).
 4. Else, increase pages (`left = mid + 1`).
-

Example:

```

makefile
CopyEdit
Input: books = [10, 20, 30, 40], students = 2
Output: Minimum max pages allocated: 60

```

 **Time Complexity: $O(N \log M)$.**

PROBLEM: 24

Problem Statement:

Given two **sorted arrays**, find their **combined median** efficiently.

Java Code:

```

java
CopyEdit
public class MedianTwoSortedArrays {
    public static void main(String[] args) {
        int[] nums1 = {1, 3};
        int[] nums2 = {2};
        System.out.println("Median: " + findMedianSortedArrays(nums1,
nums2));
    }

    public static double findMedianSortedArrays(int[] nums1, int[] nums2) {
        if (nums1.length > nums2.length) return
findMedianSortedArrays(nums2, nums1);

        int x = nums1.length, y = nums2.length;
        int low = 0, high = x;

```

```

        while (low <= high) {
            int partitionX = (low + high) / 2;
            int partitionY = (x + y + 1) / 2 - partitionX;

            int maxX = (partitionX == 0) ? Integer.MIN_VALUE :
nums1[partitionX - 1];
            int minX = (partitionX == x) ? Integer.MAX_VALUE :
nums1[partitionX];

            int maxY = (partitionY == 0) ? Integer.MIN_VALUE :
nums2[partitionY - 1];
            int minY = (partitionY == y) ? Integer.MAX_VALUE :
nums2[partitionY];

            if (maxX <= minY && maxY <= minX) {
                if ((x + y) % 2 == 0) return ((double)Math.max(maxX, maxY)
+ Math.min(minX, minY)) / 2;
                else return (double)Math.max(maxX, maxY);
            } else if (maxX > minY) high = partitionX - 1;
            else low = partitionX + 1;
        }
        throw new IllegalArgumentException();
    }
}

```

Explanation:

1. **Binary search** on the smaller array for an optimal partition.
2. **Partitions** divide both arrays into **left** (smaller) and **right** (larger).
3. Ensure $\max(\text{left}) \leq \min(\text{right})$.
4. If **even length**, return **average** of middle elements.
5. If **odd length**, return the **larger middle element**.

Example:

```

makefile
CopyEdit
Input: nums1 = [1, 3], nums2 = [2]
Output: Median: 2.0

```

 **Time Complexity: $O(\log(\min(N, M)))$.**

PROBLEM: 25

Problem Statement:

Given n stalls and c cows, place the cows in the stalls so that the **minimum distance** between any two cows is **maximized**.

Java Code:

```

java
CopyEdit
import java.util.Arrays;

```

```

import java.util.Scanner;

public class AggressiveCows {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter number of stalls: ");
        int n = scanner.nextInt();
        int[] stalls = new int[n];

        System.out.println("Enter stall positions: ");
        for (int i = 0; i < n; i++) {
            stalls[i] = scanner.nextInt();
        }

        System.out.print("Enter number of cows: ");
        int cows = scanner.nextInt();
        scanner.close();

        int result = findMaxMinDistance(stalls, cows);
        System.out.println("Largest minimum distance: " + result);
    }

    public static int findMaxMinDistance(int[] stalls, int cows) {
        Arrays.sort(stalls);
        int left = 1, right = stalls[n - 1] - stalls[0];
        int result = 0;

        while (left <= right) {
            int mid = left + (right - left) / 2;
            if (canPlaceCows(stalls, cows, mid)) {
                result = mid;
                left = mid + 1;
            } else {
                right = mid - 1;
            }
        }
        return result;
    }

    private static boolean canPlaceCows(int[] stalls, int cows, int minDist)
    {
        int count = 1, lastPlaced = stalls[0];
        for (int i = 1; i < stalls.length; i++) {
            if (stalls[i] - lastPlaced >= minDist) {
                count++;
                lastPlaced = stalls[i];
            }
            if (count == cows) return true;
        }
        return false;
    }
}

```

Explanation:

5. **Sort the stalls** in ascending order.
6. **Binary search** on possible distances.
7. Use `canPlaceCows()` to check if cows can be placed with `minDist`.
8. If yes, try a larger distance (`left = mid + 1`); otherwise, reduce (`right = mid - 1`).

Example:

makefile

CopyEdit

Input: stalls = [1, 2, 8, 4, 9], cows = 3

Output: Largest minimum distance: 3

 **Time Complexity: $O(N \log M)$** (M is max stall position).

PROBLEM: 26

Problem Statement:

Allocate n books to m students such that the **maximum pages assigned to a student** is minimized.

Java Code:

java

CopyEdit

```
import java.util.Scanner;
```

```
public class BookAllocation {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter number of books: ");
        int n = scanner.nextInt();
        int[] books = new int[n];

        System.out.println("Enter pages in books: ");
        for (int i = 0; i < n; i++) {
            books[i] = scanner.nextInt();
        }

        System.out.print("Enter number of students: ");
        int students = scanner.nextInt();
        scanner.close();

        int result = findMinimumPages(books, students);
        System.out.println("Minimum max pages allocated: " + result);
    }

    public static int findMinimumPages(int[] books, int students) {
        int left = 0, right = 0;
        for (int pages : books) {
            left = Math.max(left, pages);
            right += pages;
        }

        while (left < right) {
            int mid = left + (right - left) / 2;
            if (canAllocate(books, students, mid)) right = mid;
            else left = mid + 1;
        }
        return left;
    }
}
```

```

        private static boolean canAllocate(int[] books, int students, int
maxPages) {
            int studentCount = 1, pagesSum = 0;
            for (int pages : books) {
                if (pagesSum + pages > maxPages) {
                    studentCount++;
                    pagesSum = 0;
                }
                pagesSum += pages;
            }
            return studentCount <= students;
        }
    }
}

```

Explanation:

5. **Binary search** between `max(books)` and `sum(books)`.
6. Check if `mid` pages can be allocated using `canAllocate()`.
7. If `canAllocate()`, try a smaller value (`right = mid`).
8. Else, increase pages (`left = mid + 1`).

Example:

```

makefile
CopyEdit
Input: books = [10, 20, 30, 40], students = 2
Output: Minimum max pages allocated: 60

```

 **Time Complexity: $O(N \log M)$.**

PROBLEM: 27

Problem Statement:

Given two **sorted arrays**, find their **combined median** efficiently.

Java Code:

```

java
CopyEdit
public class MedianTwoSortedArrays {
    public static void main(String[] args) {
        int[] nums1 = {1, 3};
        int[] nums2 = {2};
        System.out.println("Median: " + findMedianSortedArrays(nums1,
nums2));
    }

    public static double findMedianSortedArrays(int[] nums1, int[] nums2) {
        if (nums1.length > nums2.length) return
findMedianSortedArrays(nums2, nums1);

```



```

    int x = nums1.length, y = nums2.length;
    int low = 0, high = x;

    while (low <= high) {
        int partitionX = (low + high) / 2;
        int partitionY = (x + y + 1) / 2 - partitionX;

        int maxX = (partitionX == 0) ? Integer.MIN_VALUE :
nums1[partitionX - 1];
        int minX = (partitionX == x) ? Integer.MAX_VALUE :
nums1[partitionX];

        int maxY = (partitionY == 0) ? Integer.MIN_VALUE :
nums2[partitionY - 1];
        int minY = (partitionY == y) ? Integer.MAX_VALUE :
nums2[partitionY];

        if (maxX <= minY && maxY <= minX) {
            if ((x + y) % 2 == 0) return ((double)Math.max(maxX, maxY)
+ Math.min(minX, minY)) / 2;
            else return (double)Math.max(maxX, maxY);
        } else if (maxX > minY) high = partitionX - 1;
        else low = partitionX + 1;
    }
    throw new IllegalArgumentException();
}
}

```

Explanation:

6. **Binary search** on the smaller array for an optimal partition.
 7. **Partitions** divide both arrays into **left** (smaller) and **right** (larger).
 8. Ensure $\max(\text{left}) \leq \min(\text{right})$.
 9. If **even length**, return **average** of middle elements.
 10. If **odd length**, return the **larger middle element**.
-

Example:

```

makefile
CopyEdit
Input: nums1 = [1, 3], nums2 = [2]
Output: Median: 2.0

```

 **Time Complexity:** $O(\log(\min(N, M)))$.

PROBLEM: 28

Problem Statement:

Given an array `nums` and an integer `m`, partition the array into `m` non-empty subarrays such that the **largest sum among these subarrays is minimized**.

Java Code:

```
java
CopyEdit
import java.util.Scanner;

public class SplitArrayLargestSum {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter number of elements: ");
        int n = scanner.nextInt();
        int[] nums = new int[n];

        System.out.println("Enter array elements: ");
        for (int i = 0; i < n; i++) {
            nums[i] = scanner.nextInt();
        }

        System.out.print("Enter number of subarrays: ");
        int m = scanner.nextInt();
        scanner.close();

        int result = splitArray(nums, m);
        System.out.println("Minimum largest sum: " + result);
    }

    public static int splitArray(int[] nums, int m) {
        int left = 0, right = 0;
        for (int num : nums) {
            left = Math.max(left, num);
            right += num;
        }

        while (left < right) {
            int mid = left + (right - left) / 2;
            if (canSplit(nums, m, mid)) right = mid;
            else left = mid + 1;
        }
        return left;
    }

    private static boolean canSplit(int[] nums, int m, int maxSum) {
        int count = 1, sum = 0;
        for (int num : nums) {
            if (sum + num > maxSum) {
                count++;
                sum = 0;
            }
            sum += num;
        }
        return count <= m;
    }
}
```

Explanation:

1. **Binary search** for the minimum possible largest sum.
2. **Check feasibility** using `canSplit()`.
3. If `canSplit()` returns `true`, search for a smaller maximum sum.

Example:

makefile

CopyEdit

Input: nums = [7, 2, 5, 10, 8], m = 2

Output: Minimum largest sum: 18

 **Time Complexity: $O(N \log S)$** (S = sum of elements).

PROBLEM: 29

Problem Statement:

Given k painters and n boards, each with a given length, assign boards to painters such that the **maximum time** taken by any painter is minimized.

Java Code:

java

CopyEdit

```
import java.util.Scanner;
```

```
public class PaintersPartition {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter number of boards: ");
        int n = scanner.nextInt();
        int[] boards = new int[n];

        System.out.println("Enter board lengths: ");
        for (int i = 0; i < n; i++) {
            boards[i] = scanner.nextInt();
        }

        System.out.print("Enter number of painters: ");
        int k = scanner.nextInt();
        scanner.close();

        int result = minTime(boards, k);
        System.out.println("Minimum time to paint all boards: " + result);
    }

    public static int minTime(int[] boards, int k) {
        int left = 0, right = 0;
        for (int board : boards) {
            left = Math.max(left, board);
            right += board;
        }

        while (left < right) {
            int mid = left + (right - left) / 2;
            if (canPaint(boards, k, mid)) right = mid;
            else left = mid + 1;
        }
    }
}
```

```

        return left;
    }

    private static boolean canPaint(int[] boards, int k, int maxTime) {
        int painters = 1, sum = 0;
        for (int board : boards) {
            if (sum + board > maxTime) {
                painters++;
                sum = 0;
            }
            sum += board;
        }
        return painters <= k;
    }
}

```

Explanation:

1. **Binary search** for the minimum possible max time.
2. **Check feasibility** using `canPaint()`.
3. Adjust search space accordingly.

Example:

```

pgsql
CopyEdit
Input: boards = [10, 20, 30, 40], painters = 2
Output: Minimum time to paint all boards: 60

```

 **Time Complexity: $O(N \log S)$.**

PROBLEM: 30

Problem Statement:

Given n gas stations at positions `stations[]`, and K additional stations to be placed, **minimize the maximum distance** between adjacent stations.

Java Code:

```

java
CopyEdit
import java.util.Arrays;
import java.util.Scanner;

public class MinimizeMaxDistance {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter number of gas stations: ");
        int n = scanner.nextInt();
        double[] stations = new double[n];
    }
}

```

```

        System.out.println("Enter gas station positions: ");
        for (int i = 0; i < n; i++) {
            stations[i] = scanner.nextDouble();
        }

        System.out.print("Enter number of new stations: ");
        int K = scanner.nextInt();
        scanner.close();

        double result = minMaxDistance(stations, K);
        System.out.println("Minimum possible max distance: " + result);
    }

    public static double minMaxDistance(double[] stations, int K) {
        Arrays.sort(stations);
        double left = 0, right = stations[stations.length - 1] -
stations[0];

        while (right - left > 1e-6) {
            double mid = (left + right) / 2;
            if (canPlace(stations, K, mid)) right = mid;
            else left = mid;
        }
        return right;
    }

    private static boolean canPlace(double[] stations, int K, double
maxDist) {
        int count = 0;
        for (int i = 1; i < stations.length; i++) {
            count += (int) ((stations[i] - stations[i - 1]) / maxDist);
        }
        return count <= K;
    }
}

```

Explanation:

1. **Binary search** on the possible max distance.
2. **Check feasibility** using `canPlace()`.
3. Adjust the search space.

Example:

```

makefile
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Input: stations = [1, 2, 3, 6, 7], K = 2
Output: Minimum possible max distance: 1.5

```

 **Time Complexity:** $O(N \log D)$ (D = max distance).

PROBLEM: 31

Problem Statement:

Find the **Kth smallest element** in two sorted arrays A[] and B[].

Java Code:

```
java
CopyEdit
public class KthElementTwoSorted {
    public static void main(String[] args) {
        int[] A = {2, 3, 6, 7, 9};
        int[] B = {1, 4, 8, 10};
        int K = 5;

        System.out.println("Kth element: " + findKthElement(A, B, K));
    }

    public static int findKthElement(int[] A, int[] B, int K) {
        if (A.length > B.length) return findKthElement(B, A, K);

        int low = Math.max(0, K - B.length), high = Math.min(K, A.length);
        while (low < high) {
            int midA = (low + high) / 2;
            int midB = K - midA;

            if (A[midA] < B[midB - 1]) low = midA + 1;
            else high = midA;
        }
        return Math.max(low > 0 ? A[low - 1] : Integer.MIN_VALUE, K - low >
0 ? B[K - low - 1] : Integer.MIN_VALUE);
    }
}
```

Explanation:

1. **Binary search** on A[] to find the **Kth element**.
 2. Adjust partitions accordingly.
-

Example:

```
makefile
CopyEdit
Input: A = [2, 3, 6, 7, 9], B = [1, 4, 8, 10], K = 5
Output: Kth element: 6
```

 **Time Complexity: $O(\log \min(N, M))$.**

PROBLEM: 32

Problem Statement:

Given a binary matrix `mat[][]` of size `N x M`, where each row is sorted in **non-decreasing** order, find the **row index** with the **maximum number of 1's**. If multiple rows have the same maximum count, return the index of the first such row.

Java Code:

```
public class RowWithMaxOnes {
    public static void main(String[] args) {
        int[][] mat = {
            {0, 0, 1, 1},
            {0, 1, 1, 1},
            {0, 0, 0, 1},
            {1, 1, 1, 1}
        };

        System.out.println("Row with maximum 1s: " + rowWithMax1s(mat));
    }

    public static int rowWithMax1s(int[][] mat) {
        int maxRowIndex = -1, maxCount = 0;
        int cols = mat[0].length;

        for (int i = 0; i < mat.length; i++) {
            int onesCount = cols - firstOneIndex(mat[i]);
            if (onesCount > maxCount) {
                maxCount = onesCount;
                maxRowIndex = i;
            }
        }
        return maxRowIndex;
    }

    private static int firstOneIndex(int[] row) {
        int left = 0, right = row.length - 1;
        while (left <= right) {
            int mid = left + (right - left) / 2;
            if (row[mid] == 1) right = mid - 1;
            else left = mid + 1;
        }
        return left;
    }
}
```

Explanation:

1. **Use binary search** to find the first occurrence of 1 in each row.
2. Compute the **number of 1's** in that row.
3. **Track the row** with the highest count.

Example:

Input:
`mat = [`

```
[0, 0, 1, 1],
[0, 1, 1, 1],
[0, 0, 0, 1],
[1, 1, 1, 1]
]
Output: Row with maximum 1s: 3
```

 **Time Complexity: $O(N \log M)$.**

PROBLEM: 33

Problem Statement:

Given an $N \times M$ matrix where:

- Each row is sorted from **left to right**.
 - The first integer of each row is **greater than the last integer of the previous row**.
Determine whether a target integer exists in the matrix.
-

Java Code:

```
public class Search2DMatrix {
    public static void main(String[] args) {
        int[][] matrix = {
            {1, 3, 5, 7},
            {10, 11, 16, 20},
            {23, 30, 34, 60}
        };
        int target = 16;

        System.out.println("Target found: " + searchMatrix(matrix, target));
    }

    public static boolean searchMatrix(int[][] matrix, int target) {
        int rows = matrix.length, cols = matrix[0].length;
        int left = 0, right = rows * cols - 1;

        while (left <= right) {
            int mid = left + (right - left) / 2;
            int midValue = matrix[mid / cols][mid % cols];

            if (midValue == target) return true;
            else if (midValue < target) left = mid + 1;
            else right = mid - 1;
        }
        return false;
    }
}
```

Explanation:

1. **Flatten the matrix** into a **1D array** using indexing.
2. **Apply binary search** to locate the target.

Example:

```
Input: matrix = [
    [1, 3, 5, 7],
    [10, 11, 16, 20],
    [23, 30, 34, 60]
], target = 16
Output: Target found: true
```

 **Time Complexity: $O(\log(N*M))$.**

PROBLEM: 34

Problem Statement:

Given an $N \times M$ matrix where:

- Each row is sorted from **left to right**.
 - Each column is sorted from **top to bottom**.
- Determine whether a target integer exists in the matrix.

Java Code:

```
public class SearchSortedMatrix {
    public static void main(String[] args) {
        int[][] matrix = {
            {1, 4, 7, 11},
            {2, 5, 8, 12},
            {3, 6, 9, 16},
            {10, 13, 14, 17}
        };
        int target = 5;

        System.out.println("Target found: " + searchMatrix(matrix, target));
    }

    public static boolean searchMatrix(int[][] matrix, int target) {
        int row = 0, col = matrix[0].length - 1;

        while (row < matrix.length && col >= 0) {
            if (matrix[row][col] == target) return true;
            else if (matrix[row][col] < target) row++;
            else col--;
        }
        return false;
    }
}
```

```
}  
}
```

Explanation:

1. **Start from the top-right corner.**
2. **Move left if the current value is greater** than the target.
3. **Move down if the current value is smaller** than the target.

Example:

```
Input: matrix = [  
    [1, 4, 7, 11],  
    [2, 5, 8, 12],  
    [3, 6, 9, 16],  
    [10, 13, 14, 17]  
], target = 5  
Output: Target found: true
```

 **Time Complexity: $O(N + M)$.**

PROBLEM: 35

Problem Statement:

A **peak element** in a 2D matrix is an element that is **greater than all of its 4 neighbors**. Given an $N \times M$ matrix, find **any** peak element.

Java Code:

```
public class FindPeakElement2D {  
    public static void main(String[] args) {  
        int[][] matrix = {  
            {10, 20, 15},  
            {21, 30, 14},  
            {7, 16, 32}  
        };  
  
        int[] peak = findPeakElement(matrix);  
        System.out.println("Peak Element found at: (" + peak[0] + ", " +  
peak[1] + ")");  
    }  
  
    public static int[] findPeakElement(int[][] matrix) {  
        int rows = matrix.length, cols = matrix[0].length;  
        int left = 0, right = cols - 1;  
  
        while (left <= right) {  
            int mid = left + (right - left) / 2;  
            int rowMaxIndex = getMaxIndex(matrix, mid);
```

```

        int leftValue = (mid > 0) ? matrix[rowMaxIndex][mid - 1] :
Integer.MIN_VALUE;
        int rightValue = (mid < cols - 1) ? matrix[rowMaxIndex][mid +
1] : Integer.MIN_VALUE;

        if (matrix[rowMaxIndex][mid] > leftValue &&
matrix[rowMaxIndex][mid] > rightValue) {
            return new int[]{rowMaxIndex, mid};
        } else if (matrix[rowMaxIndex][mid] < leftValue) right = mid -
1;
        else left = mid + 1;
    }
    return new int[]{-1, -1};
}

private static int getMaxIndex(int[][] matrix, int col) {
    int maxIndex = 0;
    for (int i = 1; i < matrix.length; i++) {
        if (matrix[i][col] > matrix[maxIndex][col]) maxIndex = i;
    }
    return maxIndex;
}
}

```

Explanation:

1. Use **binary search on columns**.
2. Find the **row-wise max in the middle column**.
3. **Adjust search space** accordingly.

 **Time Complexity: $O(N \log M)$.**

```

    }

    public static void moveZeros(int[] arr) {
        int nonZeroIndex = 0;
        for (int i = 0; i < arr.length; i++) {
            if (arr[i] != 0) {
                arr[nonZeroIndex++] = arr[i];
            }
        }
        while (nonZeroIndex < arr.length) {
            arr[nonZeroIndex++] = 0;
        }
    }
}

```

PROBLEM: 36

Problem Statement:

Given an $N \times M$ **row-wise sorted** matrix, find its **median**.

- The median is the middle element when all elements are sorted.
- If the total number of elements is odd, return the middle element.
- If the total number of elements is even, return the smaller of the two middle elements.

Java Code:

```
java
CopyEdit
import java.util.Arrays;

public class MatrixMedian {
    public static void main(String[] args) {
        int[][] matrix = {
            {1, 3, 5},
            {2, 6, 9},
            {3, 6, 9}
        };

        System.out.println("Median of matrix: " + findMedian(matrix));
    }

    public static int findMedian(int[][] matrix) {
        int rows = matrix.length, cols = matrix[0].length;
        int min = Integer.MAX_VALUE, max = Integer.MIN_VALUE;

        // Find the global min and max in the matrix
        for (int i = 0; i < rows; i++) {
            min = Math.min(min, matrix[i][0]); // First element of each row
            max = Math.max(max, matrix[i][cols - 1]); // Last element of
each row
        }

        int desiredCount = (rows * cols) / 2; // Middle index
        while (min < max) {
            int mid = min + (max - min) / 2;
            int count = countLessOrEqual(matrix, mid);

            if (count <= desiredCount) min = mid + 1;
            else max = mid;
        }
        return min;
    }

    private static int countLessOrEqual(int[][] matrix, int num) {
        int count = 0;
        for (int[] row : matrix) {
            count += countSmallerEqual(row, num);
        }
        return count;
    }

    private static int countSmallerEqual(int[] row, int num) {
        int left = 0, right = row.length - 1;
        while (left <= right) {
            int mid = left + (right - left) / 2;
```

```

        if (row[mid] <= num) left = mid + 1;
        else right = mid - 1;
    }
    return left; // Position of first greater element = count of ≤ num
elements
    }
}

```

Explanation:

1. **Find the min & max** values in the matrix.
 2. **Use binary search** on the range [min, max].
 3. **Count numbers ≤ mid** in the matrix using binary search.
 4. If count is ≤ **median index**, search the **right half**; otherwise, search the **left half**.
-

 **Time Complexity: $O(N \log M \log \text{max-min})$.**

 **Space Complexity: $O(1)$** (No extra space used).

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